

ufactured by KBM. The missile has a maximum operational range of 5.2km and a peak speed of 800m/s. It has three variants, namely, 9K310 Igla-1E (NATO designation SA-16 Gimlet), 9K38 Igla (NATO designation SA-18 Grouse) and 9K38 Igla-S (NATO SA-24). Igla missile system was inducted into service in 1981, and its different variants developed over the years continue to be in service till date.

9K310 Igla-1 with its 9M313 missile uses liquid nitrogen cooled indium-antimonide IR seeker head. 9K38 Igla with its 9M39 missile was inducted into service in 1983. It used liquid nitrogen cooled indium-antimonide and uncooled lead-sulphide IR seeker head that has higher sensitivity and improved resistance to countermeasures and jamming.

Igla-S (SA-24 Grinch) shown in Fig. 17 is the latest generation of portable air defence missile system designed to target visible aerial platforms, such as helicopters, tactical aircraft, unmanned aerial vehicles and cruise missiles. It is an improvement over earlier SA-16 and SA-18 versions. It employs a dual-band IR seeker and has a maximum engagement range of 6km as compared to 5.2km in the case of SA-16 and SA-18. It also uses a heavier warhead, which allows it to destroy the target even if it misses the target by 1.5m.

Strela family of missiles is man-portable surface-to-air missiles that use passive IR homing guidance and a high explosive warhead. Different members of the family are 9K31 Strela-1, 9K32 Strela-2, 9K34 Strela-3 and 9K37 Strela-10. Strela-1 is commonly known by its NATO designation SA-9 Gaskin and is a short-range, low-altitude, self-propelled SAM-carrying system based on the BRDM-2 chassis, an amphibious patrol car mounting two pairs of ready-to-fire 9M31 missiles. The missile has a maximum operational range of 4.2km and speed of 1.8Mach. It uses a lead-sulphide IR seeker.

9K32 Strela-2 (NATO designation SA-7 Grail) is a man-portable, should-



Fig. 17: 9K38 Igla-S (SA-24 Grinch) missile system

der-fired, low-altitude surface-to-air missile system with maximum firing range of 3700m (Strela-2) and 4200m (Strela-2M). It was inducted into service in 1968. SA-7 Grail is a tail-chase missile system whose efficacy depends on its ability to lock onto the heat source of low-flying fixed and rotary-wing aircraft. The simple IR seeker mechanism of the missile is easily prone to simple countermeasures and environmental effects.

9K34 Strela-3 (NATO designation SA-14 Gremlin) was developed to overcome the shortcomings of its predecessor Strela-2. 9M36-1 missile of SA-14 Gremlin used a new IR homing seeker that was less vulnerable to jamming and decoy flares as compared to SA-7 Grail. It has a maximum operational range of 4.5km and average supersonic speed of 410m/s.

9K35 Strela-10 (NATO designation SA-13 Gopher) was designed to replace Strela-1. It was an improvement on Strela-1 and had an effective firing range of 5km. The 9M37 missiles of Strela-10 used a higher quality IR seeker than was used in 9M31 missiles of Strela-1 system.

Air-to-air missiles. Air-to-air missiles are broadly grouped as short-range air-to-air missiles (SRAAM), also sometimes known as within-visual-range air-to-air missiles (WVRAAM) or dogfight missiles, and medium-range air-to-air missiles (MRAAM) and long-range air-to-air missiles (LRAAM). The second group of missiles is also known as beyond-visual-range air-to-air missiles (BVRAAM).

While missiles of the first group

that have engagement ranges up to 30km are usually heat-seeking missiles, missiles of the second group largely employ radar guidance. In the case of long-range missiles, IR signatures of the target aircraft would be too weak for the detector to be able to track the target. The short-range IR-guided air-to-air missiles have seen five generations of development. These developments have mainly been in IR seeker technologies and to some extent in digital signal processing.

The first generation of these missiles used IR seekers that had a field-of-view of 30° and the attack aircraft needed to position themselves behind the target aircraft during attack. The target in that case could easily move out of the seeker's field-of-view with a simple manoeuvre. Second-generation missiles used IR seekers with field-of-view of 45°. Third-generation missiles were all-aspect missiles, which meant that, the attack aircraft did not have to position itself behind the target aircraft.

Fourth-generation missiles used advanced seekers that had higher resistance to IR countermeasures and increased field-of-view of 120°, giving these higher off-bore sight capability of 60°. Fifth-generation missiles used IIR seekers and more powerful digital signal processing, which gave these higher immunity to IR countermeasures like flares, greater sensitivity and ability to hit vulnerable points on the target.

Some of the well-known contemporary air-to-air missiles include IRIS-T of Germany, Vypel R-73 of Russia, MBDA MICA-IR of France, AIM-132 ASRAAM of Great Britain, AIM-9X Sidewinder and Python-5 of Israel.

IRIS-T is a short-range air-to-air missile manufactured by Diehl BGT Defence. It employs an IIR seeker. It has a maximum speed of 3Mach and operational range of approximately 25km. It was developed to replace AIM-9 Sidewinder missile. IRIS-T has a higher resistance to IR coun-